

Lines and Queries I

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to efficiently process the following types of queries:

1. Add a line $ax + b$
2. Find the maximum point in any line at position x

Input

The first line has an integer n : the number of queries.

The following n lines describe the queries. The format of each line is either "1 a b " or "2 x ".

You may assume that the first query is of type 1.

Output

Print the answer for each query of type 2.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq a, b \leq 10^9$
- $0 \leq x \leq 10^5$

Example

Input	Output
6 1 1 2 2 1 2 3 1 0 4 2 1 2 3	3 5 4 5